

Stochastic Variational Inference (SVI) Assignment — Solutions

ELG 5218 — Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Context (given). We observe data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ generated by

$$y_i = w x_i + b + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, s).$$

We fit the Bayesian linear regression model

$$\begin{aligned} w &\sim \mathcal{N}(0, 10^2), & b &\sim \mathcal{N}(0, 10^2), & s &\sim \text{HalfCauchy}(10), \\ y_i &| w, b, s, x_i \sim \mathcal{N}(w x_i + b, s). \end{aligned}$$

We approximate the posterior $p(\theta | \mathcal{D})$ over $\theta = (w, b, s)$ using NumPyro SVI. Unless stated otherwise, use $N = 60$ data points generated with $w_{\text{true}} = 0.5$, $b_{\text{true}} = -0.7$, $s_{\text{true}} = 1.0$, $x_i \sim \mathcal{N}(0, 5^2)$ (sorted), and $y_i = w_{\text{true}} x_i + b_{\text{true}} + \varepsilon_i$.

Conceptual Understanding (SVI & Variational Inference)

1.1 VI vs SVI (scaling).

- (a) **Solution:** Full-batch variational inference (VI) maximizes the ELBO using gradients computed on the *entire* dataset, so each optimization step uses all N observations. Stochastic variational inference (SVI) maximizes the *same* ELBO objective, but replaces full-data gradients with *stochastic* (minibatch) gradient estimates and updates parameters using SGD-like steps. As a result, SVI has much cheaper updates and can scale to very large N , at the cost of noisier gradients and typically more iterations. In practice SVI uses learning-rate schedules (and sometimes multiple Monte Carlo particles) to control the variance of stochastic gradients.
- (b) **Solution:** With conditional independence,

$$\log p(\mathcal{D} | \theta) = \sum_{i=1}^N \log p(y_i | x_i, \theta).$$

The ELBO contains the expected log-likelihood term $\sum_{i=1}^N \mathbb{E}_{q_\nu}[\log p(y_i | x_i, \theta)]$, so evaluating it (or its gradient) requires summing N contributions; therefore a single gradient step costs $\Theta(N)$ work (up to the cost of computing each term).

- (c) **Solution:** Let $\mathcal{B}$ be a minibatch sampled uniformly from $\{1, \dots, N\}$ with $|\mathcal{B}| = M$. An unbiased estimator of the log-likelihood sum is

$$\sum_{i=1}^N \log p(y_i | x_i, \theta) \approx \frac{N}{M} \sum_{i \in \mathcal{B}} \log p(y_i | x_i, \theta),$$

and correspondingly for the ELBO term

$$\sum_{i=1}^N \mathbb{E}_{q_\nu}[\log p(y_i | x_i, \theta)] \approx \frac{N}{M} \sum_{i \in \mathcal{B}} \mathbb{E}_{q_\nu}[\log p(y_i | x_i, \theta)].$$

- (d) **Solution:** Minibatching trades *computation per step* for *higher-variance (noisier) gradient estimates*, which can slow or destabilize convergence unless the optimizer/learning rate is tuned.

1.2 ELBO and KL equivalence (proof).

- (a) **Solution:** For a variational family $\{q_\nu(\theta)\}$, the ELBO is

$$\mathcal{L}(\nu) = \mathbb{E}_{q_\nu(\theta)}[\log p(\mathcal{D}, \theta) - \log q_\nu(\theta)] = \mathbb{E}_{q_\nu}[\log p(\mathcal{D} | \theta)] - \text{KL}(q_\nu(\theta) \| p(\theta)).$$

- (b) **Solution:** Start from Bayes' rule $p(\theta | \mathcal{D}) = \frac{p(\mathcal{D}, \theta)}{p(\mathcal{D})}$, i.e. $\log p(\mathcal{D}, \theta) = \log p(\theta | \mathcal{D}) + \log p(\mathcal{D})$. Plug this into the ELBO:

$$\begin{aligned} \mathcal{L}(\nu) &= \mathbb{E}_{q_\nu}[\log p(\theta | \mathcal{D}) + \log p(\mathcal{D}) - \log q_\nu(\theta)] \\ &= \log p(\mathcal{D}) - \mathbb{E}_{q_\nu}[\log q_\nu(\theta) - \log p(\theta | \mathcal{D})] \\ &= \log p(\mathcal{D}) - \text{KL}(q_\nu(\theta) \| p(\theta | \mathcal{D})). \end{aligned}$$

Rearranging gives

$$\log p(\mathcal{D}) = \mathcal{L}(\nu) + \text{KL}(q_\nu(\theta) \| p(\theta | \mathcal{D})).$$

- (c) **Solution:** $\log p(\mathcal{D})$ does not depend on ν , so maximizing $\mathcal{L}(\nu)$ is equivalent to minimizing $\text{KL}(q_\nu(\theta) \| p(\theta | \mathcal{D}))$.

Results Interpretation & Diagnostics

2.1 Loss/ELBO curve: concrete patterns.

- (a) **Solution:** A “healthy” SVI run typically shows an overall *downward trend* in the (negative) ELBO / loss, eventually flattening to a stable plateau (convergence). Small stochastic oscillations are normal, especially with minibatches.
- (b) **Solution:** If the curve is highly noisy or oscillatory without improvement, common causes are: learning rate too large, minibatch size too small, too few ELBO particles, or a poorly scaled model (e.g., forgetting the $\frac{N}{M}$ factor for minibatches).
- (c) **Solution:** If the loss diverges (shoots up) or becomes NaN/Inf, common causes are: invalid constrained parameters (negative scales), overly aggressive learning rate, numerical under/overflow (e.g., extreme s), or unstable gradient estimates. Remedies include lower learning rate, stronger constraints/transforms, gradient clipping, and more stable initialization.

2.2 Minibatching effects.

- (a) **Solution:** The minibatch estimator of the expected log-likelihood term is

$$\sum_{i=1}^N \mathbb{E}_q[\log p(y_i | x_i, \theta)] \approx \frac{N}{M} \sum_{i \in \mathcal{B}} \mathbb{E}_q[\log p(y_i | x_i, \theta)].$$

- (b) **Solution:** The per-step loss will be noisier (higher variance) because each update uses only M points. However, each step is cheaper, so you can often make more updates per unit wall-clock time; the average trend should still decrease if learning rates are sensible.
- (c) **Solution:** Two common stabilizers are:
- reduce / schedule the learning rate (and/or increase M),
 - increase the number of ELBO particles (more Monte Carlo samples) and/or use gradient clipping.

2.3 Parameter trace diagnostics.

- (a) **Solution:** A poor-initialization pattern is a long “burn-in” where μ_w, μ_b barely move from unreasonable values while the loss stays high, or an immediate collapse/explosion of scales (e.g., $\sigma_w \rightarrow 0$ too fast or $\sigma_w \rightarrow$ very large) in the first few epochs.
- (b) **Solution:** A suboptimal-convergence pattern is an early plateau of both the loss and the parameters at implausible values (e.g., μ_w far from the data-implied slope), often accompanied by inflated noise scale or overly diffuse variational scales.
- (c) **Solution:** Remedies include: (i) better initialization (e.g., initialize (w, b) near least-squares estimates; use `init_loc_fn`), (ii) tune optimization (lower learning rate, use decay, more steps, gradient clipping), and (iii) use a richer guide (multivariate normal or flow) if the posterior is strongly correlated.

2.4 Posterior plots: underfitting, overconfidence, mean-field failure.

- (a) **Solution:** If the posterior predictive mean line misses the trend while the predictive bands are very wide, the model is often *underfitting*: it is attributing structure to noise (inflated s) rather than learning w, b precisely.
- (b) **Solution:** If the predictive bands are extremely narrow and many observed points fall outside (especially on training data), the fit is *overconfident* (underestimates uncertainty). This can happen due to an overly restrictive guide, poor optimization, or model mismatch.
- (c) **Solution:** A mean-field failure under strong correlation shows up most clearly in a *joint* posterior plot: the true posterior is a tilted/elongated ellipse (trade-off between parameters), whereas mean-field produces axis-aligned contours, so it cannot represent the diagonal “ridge” of high posterior density.

2.5 Quantitative evaluation with known ground truth.

- (a) **Solution:** With posterior samples $\{w^{(k)}\}_{k=1}^K$, estimate the posterior mean $\bar{w} = \frac{1}{K} \sum_k w^{(k)}$ and compute

$$\widehat{\text{bias}}(w) = \bar{w} - w_{\text{true}}.$$

- (b) **Solution:** A simple uncertainty summary is the posterior standard deviation

$$\widehat{\text{sd}}(w) = \sqrt{\frac{1}{K-1} \sum_{k=1}^K (w^{(k)} - \bar{w})^2},$$

or equivalently a 95% credible interval from the empirical 2.5% and 97.5% quantiles.

- (c) **Solution:** One predictive-quality metric is the held-out log predictive density (LPPD):

$$\text{LPPD} = \sum_{(x_i, y_i) \in \mathcal{D}_{\text{test}}} \log \left(\frac{1}{K} \sum_{k=1}^K p(y_i \mid x_i, \theta^{(k)}) \right),$$

or alternatives such as RMSE of posterior predictive means and empirical coverage of predictive intervals.

2.6 Predictive bands interpretation.

- (a) **Solution:** If the predictive bands are too narrow and many observations fall outside, the predictive distribution is *miscalibrated* and overconfident. In this model that usually means the inferred noise scale s (and/or parameter uncertainty) is underestimated.
- (b) **Solution:** If the bands are too wide, the model is being overly uncertain—often due to underfitting (weak information about w, b), overly diffuse variational scales, or inflated noise s .
- (c) **Solution:** Predictive uncertainty should grow away from the data because *epistemic* uncertainty increases when extrapolating. In linear regression, for a new input $x_\star$ the posterior predictive variance has a component like $x_\star^2 \text{Var}(w) + \text{Var}(b) + 2x_\star \text{Cov}(w, b)$ (plus the noise variance), which typically increases when $|x_\star|$ is far from the region where data were observed.

Mathematical Derivations

3.1 ELBO for Bayesian linear regression.

- (a) **Solution:** The joint log density is

$$\log p(\mathcal{D}, w, b, s) = \log p(w) + \log p(b) + \log p(s) + \sum_{i=1}^N \log p(y_i \mid x_i, w, b, s).$$

- (b) **Solution:** The ELBO is the expected log joint minus the entropy term:

$$\begin{aligned} \mathcal{L}(q) &= \mathbb{E}_q[\log p(\mathcal{D}, w, b, s) - \log q(w, b, s)] \\ &= \mathbb{E}_q[\log p(w) + \log p(b) + \log p(s)] + \sum_{i=1}^N \mathbb{E}_q[\log p(y_i \mid x_i, w, b, s)] - \mathbb{E}_q[\log q(w, b, s)]. \end{aligned}$$

3.2 Log-normal noise guide and KL in log space.

(a) **Solution:** If $z \sim \mathcal{N}(\mu, \sigma^2)$ and $s = \exp(z)$, then s is LogNormal:

$$q(s) = \text{LogNormal}(\mu, \sigma^2), \quad q(s) = \frac{1}{s \sigma \sqrt{2\pi}} \exp\left(-\frac{(\log s - \mu)^2}{2\sigma^2}\right), \quad s > 0.$$

(b) **Solution:** For a Half-Cauchy prior $p(s) = \frac{2}{\pi\gamma} \frac{1}{1+(s/\gamma)^2}$ on $s > 0$ (with scale $\gamma = 10$), the induced density on $z = \log s$ is

$$p(z) = p(s = \exp z) \left| \frac{ds}{dz} \right| = \frac{2}{\pi\gamma} \frac{1}{1 + (\exp(z)/\gamma)^2} \exp(z), \quad z \in \mathbb{R}.$$

(c) **Solution:** The KL in log-space is

$$\text{KL}(q(z) \parallel p(z)) = \mathbb{E}_{q(z)}[\log q(z) - \log p(z)],$$

where $q(z) = \mathcal{N}(\mu, \sigma^2)$ and $p(z)$ is the transformed prior above. In practice this expectation is evaluated by Monte Carlo: draw $z^{(k)} \sim q$ and average $\log q(z^{(k)}) - \log p(z^{(k)})$.

3.3 Gradient of ELBO w.r.t. μ_w . Solution: Write the ELBO (suppressing other latents) as

$$\mathcal{L}(\mu_w, \sigma_w) = \mathbb{E}_{\epsilon \sim \mathcal{N}(0,1)} \left[\log p(\mathcal{D}, w, \dots) - \log q(w; \mu_w, \sigma_w) \right], \quad w = \mu_w + \sigma_w \epsilon.$$

By the reparameterization trick,

$$\nabla_{\mu_w} \mathcal{L} = \mathbb{E}_{\epsilon} \left[\nabla_{\mu_w} \left(\log p(\mathcal{D}, w, \dots) - \log q(w; \mu_w, \sigma_w) \right) \right] = \mathbb{E}_{\epsilon} \left[\frac{\partial}{\partial w} \left(\log p(\mathcal{D}, w, \dots) - \log q(w; \mu_w, \sigma_w) \right) \right],$$

since $\partial w / \partial \mu_w = 1$. For a Normal factor $q(w) = \mathcal{N}(\mu_w, \sigma_w^2)$, the μ_w -dependence of $-\log q$ cancels under this reparameterization, so a common simplified form is

$$\nabla_{\mu_w} \mathcal{L} = \mathbb{E}_{\epsilon} \left[\frac{\partial}{\partial w} \log p(\mathcal{D}, w, \dots) \right].$$

3.4 AutoDelta and MAP. Solution: Consider the variational family of point masses $q_{\phi}(\theta) = \delta(\theta - \theta_{\phi})$. Then the ELBO becomes (up to constants / degenerate entropy terms)

$$\mathcal{L}(\phi) \propto \log p(\mathcal{D}, \theta_{\phi}),$$

so maximizing the ELBO over ϕ is equivalent to maximizing the log joint density, i.e.

$$\theta_{\phi} \in \arg \max_{\theta} \log p(\theta \mid \mathcal{D}) \iff \theta_{\phi} \in \arg \max_{\theta} \log p(\mathcal{D}, \theta),$$

which is exactly MAP estimation. The approximation is degenerate because $\delta(\theta - \theta_{\phi})$ has zero posterior variance: it cannot represent uncertainty, so credible intervals and predictive bands collapse (typically leading to overconfident predictions).

ComparativeAnalysis

4.1 SVI vs HMC/NUTS (structured comparison).

- (a) **Solution:** SVI uses stochastic gradients and can operate on minibatches, so each iteration can be $O(M)$ with $M \ll N$. HMC/NUTS typically requires full-data log-density and gradients each step, so per-iteration cost is closer to $O(N)$ (plus multiple leapfrog steps per sample), making it expensive for large datasets.
- (b) **Solution:** HMC/NUTS targets the exact posterior in the limit of long runs (up to Monte Carlo error and diagnostics). SVI returns an approximation whose accuracy is limited by the variational family and by optimization; even with perfect optimization, it may have bias (e.g., underestimating variance under reverse-KL).
- (c) **Solution:** HMC/NUTS naturally recovers posterior correlations because it samples from the full joint distribution. SVI only captures dependencies if the guide can represent them (e.g., full-covariance Gaussian or flows); a mean-field guide enforces independence and cannot match correlated posteriors.
- (d) **Solution:** SVI is sensitive to optimizer choices (learning rate, minibatch size, number of particles) and to guide expressiveness, but often scales well in high dimensions. HMC can be sensitive to geometry (e.g., funnels), step-size/mass-matrix adaptation, and divergent transitions, though NUTS automates much of the tuning; when it works, it provides reliable uncertainty with diagnostic tools.

4.2 Why can AutoMultivariateNormal be better but slower?

- (a) **Solution:** A multivariate Gaussian guide can match correlated posteriors: it can represent $\text{Cov}(w, b) \neq 0$ (and other dependencies), which can improve both the fitted means and the uncertainty calibration when the true posterior is strongly correlated.
- (b) **Solution:** It can be slower because it must learn and manipulate a full covariance (typically via a Cholesky factor). For d latent dimensions this introduces $O(d^2)$ variational parameters and $O(d^3)$ linear-algebra costs (e.g., Cholesky/solves) per update, and optimization can be less stable than with a diagonal (mean-field) covariance.